

THE HERITAGE SCHOOL
FIRST TERM EXAMINATION, 2026-27
MATHEMATICS · CLASS 12 · PRACTICE PAPER 2

Maximum Marks: 80

Time Allotted: Three Hours

Instructions to Candidates

- You are allowed an additional 15 minutes for only reading the paper.
- You must NOT start writing during reading time.
- The Question Paper consists of **twenty questions** and **four sections**: Sections A, B, C and D. **All questions are compulsory.**
- **Section A** consists of very short answer questions of **1 mark** each.
- **Section B** consists of short answer questions of **2 marks** each.
- **Section C** consists of moderately long answer questions of **3 marks** each.
- **Section D** consists of long answer questions of **5 marks** each.
- Internal choices have been provided in **three questions** in each of Sections B, C and D.
- While attempting Multiple Choice Questions in Section A, you are required to write only ONE option as the answer.
- The intended marks for questions or parts of questions are given in brackets [].
- All workings, including rough work, should be done on the same page as, and adjacent to, the rest of the answer.
- Mathematical tables and graph papers are provided.

Syllabus covered: Relations and Functions; Inverse Trigonometric Functions; Matrices; Determinants; Continuity and Differentiability; Applications of Derivatives; Integrals; Application of Integrals; Linear Programming.

Not examined in this paper (held back to the Pre-Board): Differential Equations, Vectors, Three-dimensional Geometry, Probability.

SECTION A - 20 MARKS

Question 1 [20 × 1]

In sub-parts (i) to (xx), choose the correct option or answer the question as instructed.

(i) The relation R on $\mathbb{R}$ defined by $R = \{(a, b) : a \leq b^2\}$ is:

- (a) not reflexive
- (b) reflexive and symmetric
- (c) an equivalence relation
- (d) reflexive but not symmetric

(ii) The function $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^3$ is:

- (a) a bijection
- (b) one-one but not onto
- (c) onto but not one-one
- (d) neither one-one nor onto

(iii) Write the principal value of $\sin^{-1}(-1/2)$.

(iv) Find the value of $\tan^{-1}(1) + \cos^{-1}(-1/2)$.

(v) If $A =$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \text{ find } A + A'.$$

(vi) A is a square matrix of order 3 satisfying $A' = -A$. Write the value of the diagonal element a_{11} .

(vii) Find the inverse of

$$\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

(viii) If A is a 2×2 matrix with $|A| = 4$, find $|\text{adj } A|$.

(ix) Is the function

$$f(x) = \begin{cases} (x^2 - 1)/(x - 1) & x \neq 1 \\ 2 & x = 1 \end{cases}$$

continuous at $x = 1$? Justify.

(x) Differentiate $\tan^{-1}(x^2)$ with respect to x .

(xi) If $y = \log(\log x)$, find dy/dx .

(xii) If $x = a \cos \theta$ and $y = a \sin \theta$, find dy/dx .

(xiii) Find the point on the curve $y = x^2$ at which the tangent is parallel to the x -axis.

(xiv) The function $f(x) = e^{2x}$ is strictly increasing on:

- (a) $\mathbb{R}$
- (b) $(0, \infty)$ only
- (c) $(-\infty, 0)$ only
- (d) no interval

(xv) Evaluate $\int e^{3x} dx$.

(xvi) Evaluate $\int_0^{\pi/2} \cos x dx$.

(xvii) Evaluate $\int_0^1 1/(1 + x^2) dx$.

(xviii) Find the area bounded by $y = x^2$, the x -axis and the lines $x = 0$, $x = 3$.

(xix) The feasible region of an LPP is unbounded and M is the largest value of $Z = ax + by$ at a corner point. Then M is the maximum value of Z if:

- (a) the open half plane $ax + by > M$ has no point in common with the feasible region
- (b) the feasible region is contained in $ax + by > M$
- (c) M is attained at two corner points
- (d) $M > 0$

(xx) **Assertion (A):** if two rows of a determinant are identical, its value is 0.

Reason (R): interchanging two rows of a determinant changes its sign.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

SECTION B - 14 MARKS

Question 2 [2]

Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be defined by $f(n) = n^2 + n + 1$. Show that f is one-one but not onto.

Question 3 [2]

Prove that $2 \tan^{-1}(1/3) = \tan^{-1}(3/4)$.

OR

Find the value of $\sin^{-1}(\sin 5\pi/6)$.

Question 4 [2]

If $A =$

$$\begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}, \text{ show that } A^2 - 5A + 7I = O.$$

Question 5 [2]

The radius of a circle is increasing at 0.7 cm/s. Find the rate at which its circumference increases.

Question 6 [2]

If $x^2 + y^2 = 4$, find dy/dx .

OR

If $y = (\log x)^{\cos x}$, find dy/dx .

Question 7 [2]

Evaluate $\int x e^x dx$.

Question 8 [2]

Find the maximum value of $Z = 3x + 4y$ subject to $x + y \leq 4$, $x \geq 0$, $y \geq 0$.

OR

Write the corner points of the feasible region determined by $x + 2y \leq 8$, $x \geq 0$, $y \geq 0$.

SECTION C - 21 MARKS**Question 9 [3]**

Show that the relation R on $A = \{1, 2, 3, 4, 5\}$ given by $R = \{(a, b) : |a - b| \text{ is even}\}$ is an equivalence relation. Write its equivalence classes.

Question 10 [3]

Using properties of determinants, prove that

$$\begin{vmatrix} x+4 & 2x & 2x \\ 2x & x+4 & 2x \\ 2x & 2x & x+4 \end{vmatrix} = (5x+4)(4-x)^2$$

Question 11 [3]

If $y = e^{a \cos^{-1} x}$, $-1 \leq x \leq 1$, show that $(1 - x^2)(d^2y)/(dx^2) - x(dy/dx) - a^2y = 0$.

OR

Find a and b so that

$$\begin{cases} 5 & x \leq 2 \\ ax + b & 2 < x < 10 \\ 21 & x \geq 10 \end{cases}$$

is continuous everywhere.

Question 12 [3]

Find the intervals in which $f(x) = 2x^3 - 9x^2 + 12x + 15$ is (i) strictly increasing, (ii) strictly decreasing.

OR

Find the equations of the tangent and the normal to the curve $y = x^3 - 3x + 2$ at the point $(2, 4)$.

Question 13 [3]

Evaluate $\int (x+3)/(\sqrt{5-4x-x^2}) dx$.

OR

Evaluate $\int (2x+1)/((x+2)(x-3)) dx$.

Question 14 [3]

Evaluate $\int_{-5}^5 |x+2| dx$.

Question 15 [3]

Solve graphically: minimise $Z = 3x + 5y$ subject to $x + 3y \geq 3$, $x + y \geq 2$, $x \geq 0$, $y \geq 0$.

SECTION D - 25 MARKS

Question 16 [5]

Prove that $\tan^{-1}[(\sqrt{1+x} - \sqrt{1-x})/(\sqrt{1+x} + \sqrt{1-x})] = \pi/4 - (1/2)\cos^{-1}x$, for $-1/\sqrt{2} \leq x \leq 1$.

OR

Show that $f: \mathbb{R} \rightarrow \{y \in \mathbb{R} : -1 < y < 1\}$ given by $f(x) = x/(1 + |x|)$ is a bijection.

Question 17 [5]

Solve the following system of equations by the matrix method:

$$x + 2y - 3z = -4, \quad 2x + 3y + 2z = 2, \quad 3x - 3y - 4z = 11$$

Question 18 [5]

An open box with a square base is to be made out of a given quantity of cardboard of area c^2 square units. Show that the maximum volume of the box is $c^3/(6\sqrt{3})$ cubic units.

OR

A window is in the form of a rectangle surmounted by a semicircular opening, the total perimeter being 10 m. Find the dimensions that maximise the area.

Question 19 [5]

Evaluate $\int \frac{2x}{(x^2 + 1)(x^2 + 3)} dx$.

Question 20 [5]

Find the area of the region bounded by the curve $y = x^2$, the line $y = 4$ and the y-axis, in the first quadrant.

OR

Find the area of the region bounded by the parabola $x^2 = 4y$ and the line $x = 4y - 2$.

ANSWER KEY & MARKING NOTES

Practice Paper 2. **Do not read this section before attempting the paper.** Answers were checked against a computer algebra system; the notes give the decisive step rather than a full solution.

Section A

Q	Answer
1 (i)	(a) not reflexive. Take $a = 1/2$: then $a^2 = 1/4$ and $1/2 \leq 1/4$ is false.
1 (ii)	(a) a bijection. x^3 is strictly increasing, so one-one; and every real has a real cube root, so onto.
1 (iii)	$-\pi/6$. The range of $\sin^{-1}$ is $[-\pi/2, \pi/2]$ and $\sin(-\pi/6) = -1/2$.
1 (iv)	$(11\pi)/12$. $\tan^{-1} 1 = \pi/4$ and $\cos^{-1}(-1/2) = 2\pi/3$, and $\pi/4 + 2\pi/3 = (11\pi)/12$.
1 (v)	$A + A' = [[2, 5], [5, 8]]$. (The sum of a matrix and its transpose is always symmetric.)
1 (vi)	0. Skew symmetry forces $a_{ii} = -a_{ii}$ for every i , so every diagonal entry is 0.
1 (vii)	$[[1/2, 0], [0, 1/3]]$. For a diagonal matrix the inverse is diagonal with the reciprocals.
1 (viii)	$ \text{adj } A = A ^{n-1} = 4^1 = 4$.
1 (ix)	Yes. For $x \neq 1$ the expression is $x + 1$, whose limit at $x = 1$ is 2, and $f(1) = 2$.
1 (x)	$2x/(1 + x^4)$.
1 (xi)	$1/(x \log x)$.
1 (xii)	$-\cot \theta$. $dy/dx = (a \cos \theta)/(-a \sin \theta) = -\cot \theta$.
1 (xiii)	$(0, 0)$. $dy/dx = 2x = 0$ gives $x = 0$.
1 (xiv)	(a) $\mathbb{R}$. $f(x) = 2e^{2x} > 0$ for every real x .
1 (xv)	$(e^{3x})/3 + C$.
1 (xvi)	1. $[\sin x]_0^{\pi/2} = 1 - 0$.
1 (xvii)	$\pi/4$. $[\tan^{-1} x]_0^1 = \tan^{-1} 1 - \tan^{-1} 0$.
1 (xviii)	9 square units. $\int_0^3 x^2 dx = [x^3/3]_0^3 = 9$.
1 (xix)	(a). This is the standard test for an unbounded region; without it the corner point value need not be the maximum.
1 (xx)	(a). Interchanging the two identical rows leaves the determinant unchanged but must also change its sign, so $D = -D$ and hence $D = 0$.

Section B

Q	Answer
2	One-one: for $n \geq 1$ the function is strictly increasing, so $f(n_1) = f(n_2)$ forces $n_1 = n_2$. Not onto: $f(1) = 3$ is the smallest value, so 1 and 2 have no pre-image.
3	$2 \tan^{-1} t = \tan^{-1}(2t/(1 - t^2))$ with $t = 1/3$ gives $\tan^{-1}((2/3)/(8/9)) = \tan^{-1}(3/4)$.
3 (OR)	$\pi/6$. $5\pi/6 \notin [-\pi/2, \pi/2]$, but $\sin 5\pi/6 = \sin(\pi - 5\pi/6) = \sin \pi/6$.
4	$A^2 = [[8, 5], [-5, 3]]$, $5A = [[15, 5], [-5, 10]]$ and $7I = [[7, 0], [0, 7]]$; $A^2 - 5A + 7I$ is the zero matrix.
5	$C = 2\pi r$, so $dC/dt = 2\pi(0.7) = 1.4\pi$ cm/s (≈ 4.40 cm/s).
6	Differentiating implicitly, $2x + 2y dy/dx = 0$, so $dy/dx = -x/y$.
6 (OR)	$\log y = \cos x \cdot \log(\log x)$. Differentiating, $(1/y)(dy/dx) = -\sin x \cdot \log(\log x) + (\cos x)/(x \log x)$, so $dy/dx = (\log x)^{\cos x} [-\sin x \log(\log x) + (\cos x)/(x \log x)]$.
7	By parts with $u = x$: $x e^x - \int e^x dx = \mathbf{e^x(x - 1) + C}$.
8	Corner points $(0,0)$, $(4,0)$, $(0,4)$ give $Z = 0, 12, 16$. Maximum $Z = 16$ at $(0, 4)$.
8 (OR)	$(0, 0)$, $(8, 0)$ and $(0, 4)$.

Section C

Q	Answer
9	Reflexive: $ a - a = 0$ is even. Symmetric: $ a - b = b - a $. Transitive: if $a - b$ and $b - c$ are both even so is their sum $a - c$. Two numbers are related exactly when they have the same parity. Classes: {1, 3, 5} and {2, 4}.
10	Apply $C_1 \rightarrow C_1 + C_2 + C_3$: every entry of the first column becomes $5x + 4$. Take it out, then use $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$ to obtain two factors of $(4 - x)$. Verified symbolically.
11	$dy/dx = (-ay)/(\sqrt{1 - x^2})$, so $\sqrt{1 - x^2} dy/dx = -ay$. Squaring and differentiating once more removes the radical and gives the result. Verified symbolically.
11 (OR)	Continuity at $x = 2$: $2a + b = 5$. At $x = 10$: $10a + b = 21$. Subtracting, $8a = 16$, so $a = 2$ and $b = 1$.
12	$f(x) = 6x^2 - 18x + 12 = 6(x - 1)(x - 2)$. Increasing on $(-\infty, 1)$ and $(2, \infty)$; decreasing on $(1, 2)$.
12 (OR)	$dy/dx = 3x^2 - 3 = 9$ at $x = 2$, and the point does lie on the curve ($8 - 6 + 2 = 4$). Tangent: $y = 9x - 14$. Normal: $x + 9y = 38$ (slope $-1/9$).
13	Write $x + 3 = -(1/2)(-4 - 2x) + 1$ and complete the square, $5 - 4x - x^2 = 9 - (x + 2)^2$. The two parts give $-\sqrt{5 - 4x - x^2} + \sin^{-1}((x + 2)/3) + C$.
13 (OR)	Partial fractions give $(3/5)/(x + 2) + (7/5)/(x - 3)$, so the integral is $3/5 \log x + 2 + 7/5 \log x - 3 + C$.
14	Split at $x = -2$. $\int_{-5}^{-2} -(x + 2) dx = 9/2$ and $\int_{-2}^5 (x + 2) dx = 49/2$, so the value is 29 .
15	The region is unbounded. Corner points $(3, 0)$, $(3/2, 1/2)$ and $(0, 2)$ give $Z = 9, 7$ and 10 . Since $3x + 5y < 7$ has no point in common with the region, minimum $Z = 7$ at $(3/2, 1/2)$.

Section D

Q	Answer
16	Put $x = \cos 2\theta$, so $\theta = (1/2)\cos^{-1}x$. Then $\sqrt{1 + x} = \sqrt{2} \cos \theta$ and $\sqrt{1 - x} = \sqrt{2} \sin \theta$, and the argument becomes $(\cos \theta - \sin \theta)/(\cos \theta + \sin \theta) = \tan(\pi/4 - \theta)$. Hence the value is $\pi/4 - \theta = \mathbf{\pi/4 - (1/2)\cos^{-1}x}$.
16 (OR)	For $x \geq 0$, $f(x) = x/(1 + x) \in [0, 1)$; for $x < 0$, $f(x) = x/(1 - x) \in (-1, 0)$. f is strictly increasing on each piece and across 0, hence one-one, and its range is exactly $(-1, 1)$. Inverse: $f^{-1}(y) = y/(1 - y)$.
17	$A = [[1, 2, -3], [2, 3, 2], [3, -3, -4]]$ has $ A = 67 \neq 0$, so $X = A^{-1}B$ gives $x = 3, y = -2, z = 1$. Check: $3 - 4 - 3 = -4 \checkmark$, $6 - 6 + 2 = 2 \checkmark$, $9 + 6 - 4 = 11 \checkmark$.
18	Let the base be $x \times x$ and the height h . Then $x^2 + 4xh = c^2$, so $h = (c^2 - x^2)/4x$ and $V = x^2h = (x(c^2 - x^2))/4$. $dV/dx = (c^2 - 3x^2)/4 = 0$ gives $x = c/\sqrt{3}$, and V then equals $(\sqrt{3} c^3)/18 = \mathbf{c^3/(6\sqrt{3})}$. The second derivative is negative there.
18 (OR)	With semicircle radius r and rectangle height h , $2r + 2h + \pi r = 10$ and $A = 10r - 2r^2 - (\pi r^2)/2$. Setting $dA/dr = 0$ gives $r = 10/(\pi + 4)$ and then $h = r$. Radius = height = $10/(\pi + 4)$ m ≈ 1.40 m, width = $20/(\pi + 4)$ m ≈ 2.80 m, maximum area = $50/(\pi + 4)$ m² ≈ 7.00 m².
19	Put $t = x^2$, so $dt = 2x dx$ and the integral becomes $\int dt/((t + 1)(t + 3))$. Partial fractions give $(1/2)[1/(t + 1) - 1/(t + 3)]$, so the value is $(1/2) \log[(x^2 + 1)/(x^2 + 3)] + C$.
20	Integrating along the y -axis, area = $\int_0^4 \sqrt{y} dy = [(2/3)y^{3/2}]_0^4 = \mathbf{16/3 \text{ square units}}$.
20 (OR)	The curves meet where $x^2 = x + 2$, i.e. $x = -1$ and $x = 2$. Area = $\int_{-1}^2 [(x + 2)/4 - x^2/4] dx = \mathbf{9/8 \text{ square units}}$.